What this writing is, and what it costs

The gain slope is read off the GAIN STATE rather than off a height: $\bar{a}' = b(1-\bar{a})^2$ evaluated at $\hat{\bar{a}}_w$. That is the whole difference from the height writing, and everything below follows from it.

The price is structural, not a tuning failure. Because $\bar{a}'$ depends on $\hat{\bar{a}}_w$, the error recursion gains the term $-2\hat{b}(1-\hat{\bar{a}}_w)M$, so the homogeneous coefficient exceeds 1 while the stage descends. Over the canonical descent that factor accumulates to 220×, and that number is not an artefact: the true slope at the trough is 107× the slope at the seed, so the writing converts a real property of the physics into error growth. The height writing's coefficient for that path is exactly zero.

The filter prices it correctly and can do nothing else. $F_e(4,4)$ carries the term, so P_{44} grows 1.8× during motion, L grows 5.6× on y_1 and 23–39× on y_2, and the estimate leans on measurements that are unbiased and noisy. The across-seed spread is 6.4× the exogenous arm's. That is the bias-for-scatter trade, bought deliberately by an optimal filter facing a model it should not trust.

Demonstrated with a deterministic injection: seeding b at 1 instead of 8/9 puts $\hat{\bar{a}}_w[0]$ a known +0.0056 above the truth, identical across all eight seeds and carrying no random information. Fed $\bar{a}'$ from the true height the offset is erased and the across-seed band stays narrow; fed $\bar{a}'$ from the belief it is not erased and the band opens out. A common-mode model error becomes seed-dependent scatter, and the split appears only once the stage moves. Figure `formC_belief_vs_command.png`.

What the reparameterisation bought

b multiplies rather than divides (user decision, 2026-08-18). Same curve family, $b_{\text{new}} = 1/b_{\text{old}}$, so the far-field limit inverts $9/8 \to 8/9$ while contact stays exactly 1.

Every known truncation is unchanged in magnitude, for one structural reason: each is a statement about the product $\bar{a}' = b(1-\bar{a})^2$, and that product is the reparameterisation invariant. Even the relative prior width is exactly invariant, since half/mid $= (hi - lo)/(hi + lo)$ is fixed under $x \mapsto 1/x$: 1.21052 % either way.

One thing is strictly better and had not been written down. In the old form $\bar{a}' = (1-\bar{a})^2/b$ is nonlinear in b, so $\partial^2\bar{a}'/\partial b^2 = 2(1-\bar{a})^2/b^3$ and a curvature term was dropped from $F_e(4,5)$ and $H(2,5)$ every step, worth $e_b/\hat{b} = 1.21\,\%$ at the one-sigma prior. In the new form $\bar{a}'$ is exactly linear in b and that term is identically zero: the b column is now exact.

S1 — the truth, and what the model replaces

$\bar{a}_{\text{true}}$ is $D_\perp$ from `calc_correction_functions` character for character, since $\bar{a} := a/a_o = 1/c_\perp$ and $c_\perp = 1/D_\perp$.

Because $\bar{a}_{\text{true}}$ is monotone in $\bar{w}$, $\bar{w}$ can be eliminated EXACTLY and the autonomous law $\bar{a}' = b_{\text{true}}(\bar{a})(1-\bar{a})^2$ carries no approximation. The model's single approximation is replacing the function $b_{\text{true}}(\bar{a})$ by a constant. Stating it this way also dissolves the apparent conflict between the two anchors: b_{true} is a curve, its ends simply read differently, and choosing an end is a design decision.

Verified at 30 digits: $\bar{a}_{\text{true}}(1) = 0$ to 5×10^{-32}, $\bar{a}'_{\text{true}}(1) = 1$ exactly, so $b_{\text{true}}(\bar{a} = 0) = 1$; and $b_{\text{true}} \to 8/9$ FROM BELOW.

b_{true} is NOT monotone. It rises from an interior minimum 0.866978 at $\bar{w} = 2.193$ to 0.888225 at the envelope top and never reaches 8/9. The envelope range is therefore $[0.866978,\ 0.888225]$, whose lower end is an interior extremum rather than an endpoint.

Seed and prior — the seed and the width come from different things

$$\hat{b}[0] = \tfrac{8}{9}, \qquad \sqrt{P_{55}[0]} = \sup_{\bar{w}\in\mathcal{E}} \left| b_{\mathrm{true}}(\bar{w}) - \tfrac{8}{9} \right| = 0.021911$$

THE SEED is the analytic anchor because $P[0]$ is the belief at $t = 0$ and the run starts in the far field. At the seed height $\bar{w} = 22.22$ the truth is $b_{\mathrm{true}} = 0.888167$; the anchor misses it by 7.2×10^{-4}, the envelope midpoint 0.877602 by 1.06×10^{-2} — fifteen times worse, because the midpoint is dragged down by an interior minimum the run reaches twenty radii later. Measured over the eight house seeds, seeding at the midpoint costs +1.81 points of overall RMS, sd 1.52, SEM 0.54, $t = +3.38$, worse on 8/8. This is the same defect as sizing floor_a from the envelope supremum, which cost 2.9 points and was repaired the same day: both take a statistic over ground the run has not covered and use it to set a belief at $t = 0$.

THE WIDTH is a statement about the run, not about $t = 0$. b is a constant state ($b[k+1] = b[k]$, $Q_{55} = 0$), so one number must serve the whole traverse including the near wall. That is step 4 of the $P[0]$ chain in `reference/shared/param_prior_rules.md`.

KNOWN CONSEQUENCE, recorded rather than repaired. The shape-acceptance criterion compares $\sup|\theta_{\mathrm{eff}} - \theta_0|$ against $\sqrt{P[0]}$, and with $\theta_0 = 8/9$ those are the same supremum, so the ratio is identically 1.0000 and the test cannot fail. That is a defect in the TEST's denominator. Distorting the prior to give the test something to do is what mis-seeded the filter, so it is not done. Related: the tool the criterion names, `verify_shape_exponent_bound.m`, DOES NOT EXIST in this repository, so no member of this family has been through it. That is the missing acceptance artifact.

CHARTER TENSION. $\mathcal{E}$ is built from `cfg.h_bottom` and `cfg.h_init`, so $\sqrt{P_{55}[0]}$ is trajectory-dependent, while CLAUDE.md describes this family's prior as global and trajectory-independent. The anchor form of the seed is global; the width is not.

CORRECTION TO AN EARLIER READING. $b[k+1] = b[k]$ with $Q_{55} = 0$ alongside S1's varying b_{true} is NOT a self-contradiction. It is the six-step $P[0]$ chain executed correctly, with the prior serving as the container for the variation. What was genuinely missing was this citation.

Truncations written with "=" — seven, none previously declared

Magnitudes are on the production scenario, $R = 2.25\,\mu$m, $\Delta t = 1/1600$, $\lambda_c = 0.7$, 1 Hz $\times$ 2.5 μm about $h = 7\,\mu$m, $\mathcal{E} = [1.9,\ 23.222]$.

1. S3's $\bar{a}_w[k+1] = \bar{a}_w + \bar{a}'_{\mathrm{true}}\Sigma$ is forward Euler for $\bar{a}(\bar{w} + \Sigma)$. Dropped: $\frac{1}{2}\bar{a}''\Sigma^2$. Per-step relative error $b(1-\bar{a})|\Sigma| = 0.200\,\%$ at the trough, $0.133\,\%$ over the cycle; accumulated $+5.90 \times 10^{-4}$ per oscillation cycle. $\bar{a}'' < 0$ everywhere and the term goes as Σ^2, so it does NOT change sign with direction of travel: a closed oscillation does not cancel it and the estimator runs high. This is the project's quadrature ratchet.

2. S6's $e_{\bar{a}_w}$ recursion is written as an identity but drops $+\hat{b}\,e^2M$, which no matrix F_e can carry. Size against the retained $-2\hat{b}(1-\hat{\bar{a}}_w)eM$ is $e/(2(1-\hat{\bar{a}}_w)) \approx 0.31\,\%$. The same expansion also drops the cross term $-2(1-\hat{\bar{a}}_w)e_{\bar{a}}e_bM$, about $1.25\,\%$ of the retained e_b column.

3. S7's back-off uses the COMMANDED $\nabla_d\bar{w}_d$; the actual d-step change is $\nabla_d\bar{w}_d - \delta\bar{w}_3[k] + \delta\bar{w}_1[k]$. So $H(2,1) = -\bar{a}'$ and $H(2,3) = +\bar{a}'$ are MISSING COLUMNS, not small coefficients. At this project's reported z tracking errors they are 25–50 % of the retained term, and $H(2,1)$ puts y_2 on the same column y_1 measures — the channel the y_1-versus-y_2 decomposition found dominant.

4. The same back-off is itself forward Euler over a $d = 2$ gap, carrying four times the single-step quadrature error: $+8.56 \times 10^{-6}$ per application at the trough, landing as a one-signed bias on the y_2 innovation rather than integrating into the state. Independent of item 3; fixing one does not fix the other.

5. S8's $q_3 = -\bar{a}_w\bar{f}_T$ dropped the MA(2) memory and the $(1-\lambda_c)n_w$ share of ε_w that S3 itself defines. $\mathrm{Var}(\varepsilon_w)$ was understated $1.18\times$ and its DC power $2.17\times$ — the same factor the formB MA(2) work landed on. The code was right and the document wrong. REPAIRED 2026-08-30: S3, S4, S5, S6 and S8 now carry the two memory states m_1, m_2 and the rank-2 Q that production runs (`ma2_aug = true`); the white-container arm's full variance is written out beside it. Measured on the truth chain (V3, ledger L44): realised $\mathrm{Var}(\varepsilon_w)$ over the truncated $\kappa_T\bar{a}$ is 1.17–1.21 against the 1.18–1.23 this item predicted, and 0.93–1.01 against the full container.

6. $P_{44}[0]$ adds the $\bar{w}_s$ and b contributions as if the two priors were independent, while $P_{45}[0]$ on the next line asserts a nonzero correlation between b and $\bar{a}$. The independence is undeclared.

7. S8 mixed $\bar{a}_w$ and $\hat{\bar{a}}_w$: Q_{33}, R_2 and $\sigma^2_{\delta\bar{w}_r}$ were written at the TRUE gain while Q_{34} and Q_{44} used the estimate and the code uses the estimate throughout. For a document whose charter constraint is run-time c-free evaluation at the estimate, that was not a leavable typo. REPAIRED 2026-08-30: S8 is at $\hat{\bar{a}}_w$ throughout; the one true-gain line left is the definition $\sigma^2_{wT} = \kappa_T\bar{a}_w$, followed by the arrow to its estimate.

An assumption violation the truncations hide

ε_w contains $-(1-\lambda_c)n_w[k]$ (sign corrected 2026-08-30 from the control law: the delayed measurement enters the tracking error through $-(1-\lambda_c)(\bar{w}-\hat{w})$ and $\bar{w}-\hat{w}$ carries $-n_w$; the code header had the minus already) and y_1's noise IS $n_w[k]$, so $E[\mathbf{q}\mathbf{v}^\top] \neq 0$, which the standard Kalman filter this document invokes forbids, and $R = \mathrm{diag}(R_1, R_2)$ carries no cross term. Coefficient 0.0157. Small, but structural rather than approximate — and invisible inside S8's own truncated q_3, which drops the n_w share. The truncation hides the violation, which is why it survived.

Domain, and what the file does not contain

$\bar{a}(\bar{w}) = 1 - 1/[b(\bar{w}-\bar{w}_s)]$ has a pole at $\bar{w}_s$ and is NEGATIVE for $\bar{w}-\bar{w}_s < 1/b \approx 1.125$, contradicting the main file's own $0 < \bar{a} < 1$. The condition $\bar{w}-\bar{w}_s > 1/b$ is stated nowhere; production carries it as `GAP_FLOOR` and the $\bar{a}$ clamps. Separately, $0 < \bar{a} < 1$ is open at 0 while S1's contact anchor is stated AT $\bar{a} = 0$.

P^f appears once, in $L[k]$, and is never defined. Missing with it: the covariance prediction, the covariance update, the state update — so $L[k]$ is computed and nothing consumes it — and the predicted measurement. The last is a trap rather than an omission: $\hat{y}_2$ is NONLINEAR, $\hat{y}_2 = \hat{\bar{a}}_w - \hat{b}(1-\hat{\bar{a}}_w)^2\nabla_d\bar{w}_d$, not $H\hat{\mathbf{x}}$, and production carries an explicit warning that using H there is wrong. As the file stands the only available reading is the linear one.

Previously also absent, repaired 2026-08-30: Q is now assembled as the rank-2 outer-product form with every nonzero entry listed; the MA(2) augmentation is represented $(m_1, m_2$; production slots 8 and 9, slots 6–7 inert). S4 is still titled "Process model" and still contains only the state vector.

Normalization scales (2026-08-30)

Three scales, not two. Lengths by R and the gain by $a_{\text{nom}} = \Delta t/\gamma_N$ [μm/pN] leave $\Delta\bar{w} = (a_{\text{nom}}/R)\,\bar{a}\,f$ with a dimensional coefficient $a_{\text{nom}}/R =: a_o$ [1/pN] still in the state equation: the unit of the gain contains 1/pN, which belongs to the force, and only the force can absorb it. The third scale is therefore a force, $f_R = 1/a_o = \gamma_N R/\Delta t$ (the force that moves the probe one radius in one step at far field; 153 pN on the house constants), and $\bar{f} = f/f_R = a_o f$ makes $\Delta\bar{w} = \bar{a}\bar{f}$ with coefficient one.

This is why $Q_{33} = 4k_BT\,a_o\bar{a}/R$ carries R and not R^2: a_o is the R-normalized far-field gain, already holding one $1/R$. Written with the physical gain the same quantity is $4k_BT\,a_{\text{nom}}\bar{a}/R^2$. The two forms are numerically identical; the question "R or R^2?" (asked in the lab, 2026-08-30) is a question about which a_o is meant, and the notation line of the main file now says which. $\xi = C_n\sigma^2_{nw}/(C_{dpmr}\kappa_T)$ is in units of $\bar{a}$ (C_{dpmr}, C_n are pure functions of λ_c, a_{pd}; σ^2_{nw} and κ_T are both in R^2), which is why it can be added to $\bar{a}$.

The controller crosses the unit boundary at five places and nowhere else: U0 init (`:355--360`: σ_n^2/R^2, a_o, κ_T, $a_{\text{disp}} = a_oR = a_{\text{nom}}$); U1 every step (`:996, :1013--1016, :1050`: lengths $/R$); U1$'$ (`:981`: an override gain in μm/pN, $/a_{\text{disp}}$); U3 (`:1075`: $f_d = \bar{f}_d/a_o$, the only de-normalization that reaches the plant); U4 display (`:922--937, :1517--1591`: $\times a_{\text{disp}}$, $\times R$; a legacy $\times a_o$ family gives 1/pN and must not be mixed with a_{disp}). The plant knows only γ_N, $c(\bar{w})$ and Δt. Changing the definition of a_o therefore means changing the power of R in κ_T and the division at U3 in the same edit; changing `:357` alone runs to completion with every number wrong by $R = 2.25$.

Figure axes follow the physical-gain convention: a normalized gain axis is labelled a_z/a_{nom} (or $\bar{a}_z$), never a_z/a_o, which is a length. Hats sit on normalized quantities only.

Results

Canonical scenario, 8 house seeds, paired, overall relative gain-error RMS. All three arms use the seed-local floor$_a$ and the anchored b prior.

	desc peak	osc RMS	overall	$\hat{b}$[end]
b fixed at 8/9	4.78 ± 3.27	1.68 ± 0.82	2.20 ± 1.33	0.88889
b estimated	5.33 ± 3.39	1.96 ± 1.01	2.46 ± 1.58	0.89218 ± 0.01096
$\bar{a}'$ from the TRUE height	1.21 ± 0.32	0.68 ± 0.47	1.16 ± 0.52	0.88752 ± 0.01505

Estimating b costs +0.258, sd 0.280, SEM 0.099, $t = +2.61$, better on 1/8. Consistently a little worse: b receives almost no information, so a free b is a locked b plus a noise source. The ceiling with a perfect integrand is 1.16 % against the 2.20 % achieved, and that gap is the price of reading $\bar{a}'$ at the belief.

TRACKING ERROR, the fourth figure row, is $R\,\delta\bar{w}_3$ in μm. With the sample alignment corrected the bias is zero in every phase — -0.74 nm on the descent, -1.04 ± 3.29 nm over the eight seeds in the initial hold, $t = -0.89$ — and the row carries noise only, RMS 25.08 nm against a 3.31 nm measurement noise spec.

Verification record

Thirteen relations CAS-checked, then an independent pass swept every occurrence of b cell by cell including the nested arrays inside F_e, re-derived S6's recursion from scratch, and reconfirmed both limits at 30 digits. No missed or half-converted b; the five surviving denominators are all in the seed and all correct for the multiplicative form.

The implementation was checked against the derivation by an IDENTITY rather than by agreement of numbers. $b_{\mathrm{new}} = 1/b_{\mathrm{old}}$ is exact, so a run locked at $b_{\mathrm{old}} = 1$ must reproduce one locked at $b_{\mathrm{new}} = 1$, and 9/8 must reproduce 8/9. Measured: $\hat{\hat{a}}_w$ identical to 0 and to 6.1×10^{-16} respectively. Only all eight conversion sites being right produces that.

Verification 2026-08-30 (normalization boundary, ledger L42–L45)

V1, unit-system invariance: every length re-expressed in nm (R, k_B, h_{init}, amplitude, measurement noise $\times 10^3$; $\gamma_N/10^3$) through a shadowed `physical_constants`, same seed. Twenty-seven logged fields: every dimensionless one agrees to 1.7×10^{-12}, lengths and gains scale by exactly 10^3, forces by 1. No boundary leaks R or a_o. V2, κ_T against the plant: nine ratios $\mathrm{Var}(\Delta\bar{w}_T)/(\kappa_T\bar{a})$ at $\bar{w} = 50, 3.32, 1.10$ on three axes are 0.994–1.007 ($N = 10^5$, 1σ 0.45 %); the wrong a_o would give 0.444. V3, the realised ε_w rebuilt from the truth chain against the container: 0.987/0.930/1.007/0.984 on hold, descent, oscillation, final hold. V4, the readout identity $\mathrm{Var}(\delta\bar{w}_r) = C_{dpmr}\kappa_T(\bar{a} + \xi)$: 0.975 far field, 1.095 at the band top (same direction as the unmodelled $\lambda_{\mathrm{eff}} \neq \lambda_c$ of 2026-08-26); $\xi(z) = 1.5265\times 10^{-2}$. Scripts, all under `test_script/scratch/`: `l4_norm_v1_unit_invariance.m`; `l4_norm_v2_kappaT_plant.m`; `l4_norm_v3v4_epsw_container.m`.

Open items

1. Of the seven truncations above, items 5 and 7 were repaired 2026-08-30; the other five are declared but not repaired. Item 3 is the one where the document is wrong rather than merely silent.

2. P^f and the four missing filter lines. The nonlinear $\hat{y}_2$ is the one that changes what a reader would build.

3. `verify_shape_exponent_bound.m` does not exist, so the shape acceptance this family's charter names has never been run on any member.

4. The acceptance criterion's denominator is the same supremum as its numerator when θ_0 is the anchor. The criterion needs a different denominator, not a distorted prior.

5. Production comments referencing "S5(b)" point at a 7-state version that was never written and whose header claim has been withdrawn.